Inflammation that develops at the site of infection induces the acute phase response. This generalized response is characterized by fever, changes in vascular permeability and changes in biosynthesis, metabolism and catabolism in many organs. These changes result in a rise in the concentration of certain proteins in the blood and a drop in the concentration of other proteins. These proteins that are elevated termed as acute phase proteins or acute phase reactants. These include C- reactive protein (CRP), fibrinogen and serum amyloid A protein. The concentration of CRP in the blood increases from a normal level of 1 mg / ml to as much as 1000 mg / ml during the acute phase response. It functions in clearance of nuclear material released from killed microbes and killed host cells during inflammation by binding to DNA, chromatin and histones.

10.3.2. Acquired (Adaptive) Immunity

The form of immunity that is mediated by lymphocytes and stimulated by exposure to infectious agents is adaptive immunity. It reflects the presence of a functional immune system that is capable of specifically recognizing and selectively eliminating foreign microorganisms and molecules. It is characterized by four characteristics namely antigenic specificity, diversity, immunologic memory, and self and non-self recognition.

Antigenic specificity

Immune responses are directed toward and able to distinguish between distinct antigens or small parts of macromolecular antigens. This fine specificity is attributed to lymphocyte antigen receptors that may bind to one molecule but not to another with only minor structural differences from the first. Antibodies can differentiate between two molecules that differ by only a single aminoacid.

Diversity

Diversity is the result of variability in the structure of the antigen binding sites of lymphocyte receptors for antigens. Diversity allows the adaptive immune system to specifically recognize billions of uniquely different structures on foreign antigens.

Immunologic memory

Immunologic memory is mediated by memory cells. Memory cells are clonally expanded progeny of T and B cells formed during the primary response following initial exposure to antigen. Memory cells are more easily activated than naive lymphocytes and mediate secondary response on subsequent exposure to antigen. They survive in a functionally quiescent state for many years, even after the elimination of the antigen. Due to this attribute, the immune system can confer life long immunity to many infectious agents.

Self and non-self recognition

The immune system is able to distinguish self (host) from nonself (foriegn) antigens and respond only to non-self molecules. The production of immune cells only against non- self molecules, is achieved by selection procedure like positive selection and negative

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